

Evaluating the Reliability Degradation Caused by Resonant Current in Photovoltaic Inverters With LCL Filters

Xinyue Zhang^{1b}, Member, IEEE, Jiacheng Sun^{1b}, Member, IEEE, Zhongzheng Zhou^{1b}, Member, IEEE, Zhen Kang, Member, IEEE, Wenli Yao^{1b}, Member, IEEE, Tamas Kerekes^{1b}, Senior Member, IEEE, Xiaobin Zhang, and Dao Zhou^{1b}, Senior Member, IEEE

Abstract—With the increasing deployment of photovoltaic (PV) systems, LCL -type grid-connected inverters have become prevalent due to their superior harmonic attenuation. However, conventional reliability assessments often overlook the impact of resonant currents arising from control-loop instability or parameter drift, which can significantly accelerate the degradation of key components. This paper presents a comprehensive reliability evaluation framework for LCL -type PV inverters that better reflects realistic operating conditions. An electrothermal model is developed that accounts for temperature-dependent power losses, IGBT thermal coupling, and capacitor degradation. Monte Carlo simulation and Weibull distribution are employed to derive converter lifetime under annual mission profiles. Experimental validation confirms that resonant currents introduce substantial electrical stress, and damping failure can reduce the B_{10} lifetime by up to 55.5%. Furthermore, the study establishes quantitative relationships between resonant frequency, total harmonic distortion (THD), and converter reliability, revealing that higher resonant frequencies improve capacitor lifetime. Furthermore, the integration of reliability considerations offers valuable insights into the design optimization of LCL -type PV converters.

Index Terms—Converter-level reliability, mission profile, LCL filter, resonant current, lifetime.

I. INTRODUCTION

RECENTLY, the growing utilization of renewable energy has prompted an increasing interest in LCL -type inverters.

Received 21 January 2025; revised 17 May 2025; accepted 27 August 2025. Date of publication 5 September 2025; date of current version 10 February 2026. Paper 2025-IPCC-0200.R1, presented at the 2023 IEEE Energy Conversion Congress and Exposition, Nashville, TN, USA, Oct. 29–Nov. 02, and approved for publication in the IEEE TRANSACTIONS ON INDUSTRY APPLICATIONS by the Industrial Power Converter Committee of the IEEE Industry Applications Society [DOI: 10.1109/ECCE53617.2023.10362702]. This work was supported in part by the National Natural Science Foundation of Shaanxi Province under Grant 2024JC-YBMS-419, and in part by the National Nature Science Foundation of China under Grant 52307016. (Corresponding author: Zhongzheng Zhou.)

Xinyue Zhang, Zhongzheng Zhou, Wenli Yao, and Xiaobin Zhang are with the School of Automation, Northwestern Polytechnical University, Xi'an 710072, China (e-mail: zhang_xin_yue@mail.nwpu.edu.cn; threezh@nwpu.edu.cn; yaowl@nwpu.edu.cn; zxb907@nwpu.edu.cn).

Jiacheng Sun and Zhen Kang are with Xi'an Thermal Power Research Institute Company, LTD, Xi'an 710032, China (e-mail: sunakon@mail.nwpu.edu.cn; kangzhenpri@126.com).

Tamas Kerekes and Dao Zhou are with the Department of Energy, Aalborg University, 9220 Aalborg, Denmark (e-mail: tak@et.aau.dk; zda@et.aau.dk).

Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TIA.2025.3606455>.

Digital Object Identifier 10.1109/TIA.2025.3606455

As an efficient interface to the grid, inverters require a filter to attenuate switching harmonics to ensure high power quality [1], [2], [3]. LCL filters, in comparison to L filters, offer several advantages, including enhanced harmonic attenuation and reduced weight and size [4], [5]. Solar PV is rapidly becoming the leading technology in the renewable energy field [6]. Reducing the cost of PV energy has become a necessary prerequisite for improving the competitiveness of PV [7]. In order to achieve this goal, the deployment of efficient and reliable interface converters can effectively reduce the operation and maintenance costs of PV systems. Studies indicate that around one-third of the maintenance cost of a PV system can be attributed to inverter failure [8]. This implies that reliability assessment and reliability-oriented design of PV converters in harsh conditions are critical [9], [10].

Numerous investigations into the reliability of power electronics have been carried out over the years. A survey of component-level reliability is conducted in [11], revealing that power devices and capacitors are the most vulnerable components in power electronic systems. Therefore, extensive research [12], [13], [14], [15], [16], [17], [18], [19], [20], [21] has been dedicated to evaluating the reliability of active and passive components. It is noteworthy, however, that only a limited number of studies have focused on the reliability of the converter-level [11], [22], [23], in which artificial intelligence has been applied to optimize design for power electronics system reliability [24]. Typically, capacitor manufacturers offer predictive models for estimating the lifetime of their products, considering factors like ambient temperature, ripple current, and applied voltage [25]. While these models are widely applied in engineering and academia, they often lack a robust physical foundation, relying instead on mathematical fitting methods for coefficient calculations [15]. To address this limitation, one alternative being explored is lifetime modeling based on mission profiles [6], [7], [11], [18], [20].

The LCL filter, as a third-order system, may generate resonant peaks that challenge the reliability of the PV converter [26], leading to higher maintenance expenses. Passive damping (PD) and active damping (AD) are commonly applied to suppress the resonant peak [27]. However, certain approaches are sensitive to parameter variations and uncertainties, rendering the converter's stability susceptible to slight parameter drifts. Long-term resonance, even in critical conditions, presents a potential risk

to component reliability [28], [29]. Similarly, in a weak grid scenario, variations in the grid impedance and LCL resonance frequency may induce oscillations within the converter [30]. It should be noted that despite the application of the above damping methods, non-ideal factors such as time delay, grid background harmonics, grid faults and variations in grid impedance can still significantly impact the performance and stability of the grid-connected inverter system in practical digital control. The generated resonant currents are transmitted and coupled between subsystems, increasing electrothermal stress on reliability-critical devices. Thus, the primary focus of this paper is to quantitatively analyze and compare the influence of resonant currents on the reliability of the PV inverter, especially on capacitors and power devices, which are most vulnerable to damage. It is important to highlight that this paper introduces resonance frequency and resonance level as key identifiers of factors influencing reliability, and as quantitative indicators of marginal system performance. Nevertheless, concerning capacitors, the majority of literature emphasizes evaluating the lifetime of aluminum electrolytic capacitors (Al-CAPs) on the DC side [16], [18], [31], with only a limited focus on filtered metalized polypropylene film capacitors (MPF-CAPs) on the AC side [32]. In [17], [22], the harmonic spectrum of the electrical stress on the capacitor current is explored under modulation techniques and grid conditions. Meanwhile, [32] analyzes the current spectrum utilizing an intricate double Fourier analysis. However, none of them have addressed a significant issue—the impact of the resonant current on both the thermal stress and the lifetime of the LCL filter capacitor. In the study presented in [19], the assessment of capacitor aging due to electrical stresses from reactive power is explored. Nevertheless, the analysis is constrained, primarily emphasizing the parameter shift in the equivalent series resistance (ESR). The lifetime of the MPF-CAP depends on numerous parameters [18]. Researchers have explored the ESR and variations in capacitance of capacitors [31]. Either the ESR increases or the capacitance value decreases, affecting the capacitor's overall lifetime [33]. Moreover, the impact of capacitance degradation over long-term operation is rarely discussed in the literature concerning PV converter lifetime prediction. This degradation influences the resonant frequency of LCL filters, making it essential to consider in LCL filter applications.

Typically, thermal stress has a more pronounced impact on lifetime compared to factors such as humidity and mechanical vibration [34]. The considerable disparity in expansion coefficients among the layers of the IGBT module results in thermal stresses during multiple power cycles of the semiconductor device [22]. These stresses accumulate fatigue in bonding wires, chip solder joints, and substrate solder joints, ultimately culminating in device failure [35]. While [7], [36], [37] have discussed the impact of mission profile, inverter control strategy, and topology on the lifetime of IGBTs, there is insufficient research addressing the reliability of LCL -type PV converters. Traditionally, device reliability is assessed through long-term thermal cycling, with cycles spanning from a few seconds to hours, reflecting the diverse time scales involved [21]. However, this approach fails to inherently capture the damage incurred by short-term thermal stress. In [13], [15], the study quantifies

and characterizes the lifetime damage caused by fundamental frequency thermal cycling in converters utilized in doubly-fed induction generators (DFIG) and modular multilevel converters (MMC). The prior-art discussions neglect the connection between resonant current and the reliability of the IGBTs. Therefore, the impact of resonant current on the junction temperature and lifetime of the device remains indeterminate.

To address this gap, this paper aims to quantify the effect of resonant currents generated by LCL on the reliability of PV inverters. The analysis evaluates the lifetime of critical reliability components within the converter, specifically capacitors and power devices, and introduces a quantitative metric that relates resonant frequency and harmonic content to reliability, highlighting the importance of lifetime prediction as a design guideline for PV converters. Compared with the traditional PV converter reliability assessment, some improvements are made:

- 1) This paper presents assessment methods aiming to comprehensively quantify and compare damage induced by resonant currents on critical components. This quantitative analysis not only offers valuable insights but also guides the selection and design of LCL -type PV inverter systems, emphasizing reliability considerations.
- 2) This paper addresses the thermal cross-coupling effects in IGBT modules and the capacitive degradation of filter capacitors. Unlike conventional thermal networks for IGBT modules that rely on Cauer and Foster models, the proposed approach focuses on extracting the thermal coupling parameters of neighboring chips. Notably, the study integrates capacitance degradation under long-term operating conditions into the reliability assessment, which is often overlooked in converter-level reliability analysis.
- 3) This paper proposes a quantitative index that links resonant frequency, harmonic content, and system reliability. It analyzes the impact of varying resonant frequencies and THD levels on the converter's lifetime. The results demonstrate that increasing the resonant frequency enhances system reliability. Based on these findings, a reliability-oriented design framework is proposed to guide converter design within a specified operational lifetime.

The rest of this paper is organized as follows. Section II describes a case study of a PV inverter based on the mission profile and the reliability evaluation process. Section III analyzes the electrical and thermal stresses. Section IV covers the converter reliability assessment process and the impact of total harmonic distortion content on lifetime. Lastly, Section V presents concluding remarks.

II. SYSTEM DESCRIPTION AND RELIABILITY EVALUATION PROCEDURE

A. System Description

The configuration and control structure of a smart home/building featuring a two-level single-phase PV inverter are depicted in Fig. 1, with corresponding parameters detailed in Table I. Notably, an efficient grid interface employs a full-bridge inverter. Besides, an LCL filter is employed to eliminate

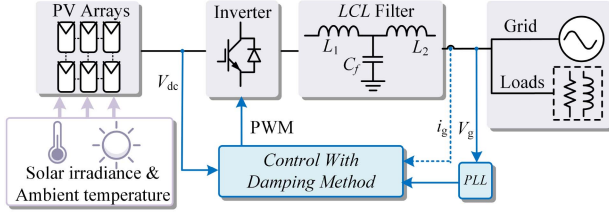


Fig. 1. General configuration and control structure of a single-phase PV inverter.

TABLE I
PARAMETERS OF THE SINGLE-PHASE PV CONVERTER

Parameters	Values
Converter-side inductor	$L_1=1.8$ mH
Grid-side inductor	$L_2=2$ mH
Filter capacitor	$C_f=15$ μ F
Grid voltage	$V_g=220$ V/50 Hz
Sample frequency	$f_s=10$ kHz
DC voltage	$V_{dc}=360$ V
Nominal power	$P_o=2.6$ kW
Resonance frequency	$f_{res}=1336$ Hz (27 th Har.)

the higher-order harmonics caused by bipolar sinusoidal pulse-width modulation (PWM). To suppress resonance, advanced damping control methods (e.g., capacitor current feedback [38], grid voltage feedforward [39]) should be employed appropriately, even if the converter satisfies the critical frequency [30], [40]. In this paper, it is achieved by regulating the grid current i_g . Additionally, a phase-locked loop (PLL) is integrated for synchronization. Despite implementing damping control, the limited robustness of the current controllers results in critical oscillations within the converter when passive component parameters deviate or grid impedance varies. These oscillations inevitably impact the reliability of key devices in the inverter system. It is important to note that this paper does not address the various causes of converter oscillations or the reliability of devices external to the inverter system. Instead, the focus is on providing guidance for evaluating lifetime estimation errors in *LCL*-type inverter systems caused by resonant currents.

Solar irradiance (SI) and ambient temperature are two parameters that are highly relevant for the mission profiles of a PV inverter. SI and ambient temperature in the mission profile are assumed to be constant during each sampling period, varying only at different sampling points [20]. The reliability assessment is applied using mission profiles obtained from the photovoltaic geographical information system (PVGIS), as depicted in Fig. 2. Based on SI, the generated power can be controlled by a proper maximum power point tracking [7].

B. Design of *LCL* Filter

The *LCL* filter is designed to mitigate high-order harmonics on the grid side, but if not properly designed, it can result in insufficient attenuation or even amplify distortion due to oscillation effects. In this paper, a standard design process is chosen, using the converter's rated power and the line frequency as input parameters [5]. In the subsequent analysis, the filter

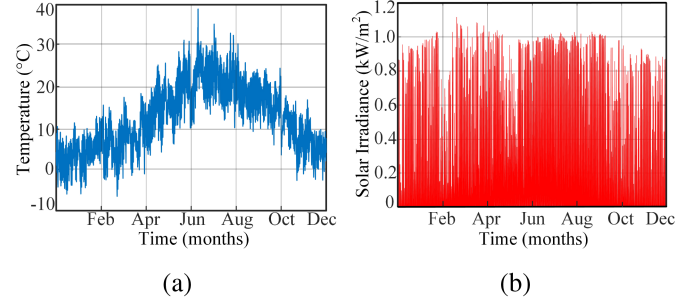


Fig. 2. One-year mission profile. (a) Ambient temperature. (b) Solar irradiance.

values are expressed as a percentage of the base values, defined by

$$Z_0 = \frac{V_g^2}{P_0} \quad (1)$$

$$C_0 = \frac{1}{\omega_g Z_0} \quad (2)$$

where V_g is the grid voltage, ω_g is the grid frequency, and P_0 is the rated power. As the filter capacitor generates reactive power, it should typically be kept below 10% of the converter's rated power:

$$C_f < 10\% C_0 \quad (3)$$

In connection with the design of the filter inductor, it is essential to account for the attenuation of switching harmonics in the converter's output current. The grid-side inductor can further reduce harmonics in the grid-side current, assuming only the switching frequency harmonic components of the converter output are taken into consideration. Based on Kirchhoff's Current Law (KCL), the relationship between the switching harmonic components $i_i(h_s)$ and $i_g(h_s)$ in the converter-side current and the grid-side current can be expressed as follows

$$\frac{i_g(h_s)}{i_i(h_s)} \approx \frac{Z_{LC}^2}{|\omega_{res}^2 - \omega_s^2|} = \frac{1}{|1 + b(1 - L_1 C \omega_s^2)|} = \alpha \quad (4)$$

where α represents the harmonic attenuation rate of the grid-side inductor, and b represents the ratio of the converter-side inductor L_1 to the grid-side inductor L_2 . With a 10% allowed ripple and 20% design attenuation, the filter parameters in Table I can be obtained, where the resonant frequency is 1336Hz (27th).

C. Critical Oscillation Analysis

While the previous section detailed a comprehensive design of *LCL* filter parameters aimed at ensuring system stability under nominal conditions, such designs cannot guarantee sustained stability over the converter's entire operational lifetime [30]. In real-world applications, filter parameters may drift over time due to factors like component aging, for example, capacitor degradation or variations in grid impedance. These gradual changes can shift the system away from its initially stable operating region, potentially driving it toward marginal stability [27].

As system parameters drift, especially the filter capacitance, the resonant frequency is significantly affected. This may lead

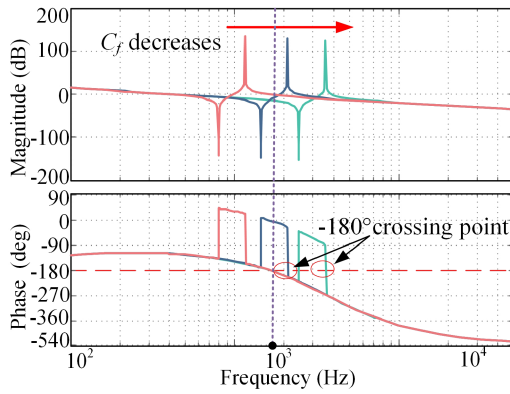


Fig. 3. Bode plot of system current loop with capacitance degradation.

to the migration of closed-loop poles toward the imaginary axis, triggering critical oscillation. Although the system may remain technically stable, it exhibits persistent oscillatory currents and pronounced resonance peaks in the frequency response [27]. As illustrated in Fig. 3, a reduction in filter capacitance results in a noticeable increase in resonant frequency as the capacitance degrades. This introduces a phase crossing at -180° behavior in the frequency response at magnitudes greater than 0 dB. According to the Nyquist stability criterion, the system transitions from a steady state to an oscillatory condition [26]. These findings clearly indicate that parameter drift can substantially degrade the system's damping and induce high-frequency oscillations.

While such oscillations may not immediately result in system instability, the high-frequency oscillatory currents exhibit a non-negligible amplitude. Resonant currents can significantly increase the electrical and thermal stress on critical components. This can accelerate failure mechanisms such as solder fatigue and dielectric loss, ultimately reducing the overall converter lifetime. Therefore, critical oscillations caused by parameter drift are not only a control challenge but also a major reliability concern. The next section will further explore the stress distribution and lifetime degradation of critical components with resonant currents.

D. Reliability Evaluation Procedure

In this paper, the lifetimes of critical components (i.e., power devices and capacitors) are evaluated with and without resonant currents based on the flowchart shown in Fig. 4. System electrical characteristics can be analyzed using PV and converter models, with a current control loop and damping method integrated into the converter model. When the damping method fails, the generated resonant currents cause fluctuations in the electrical stress of the power module and the filter capacitor. Following this, the power losses and junction/hotspot temperatures of the components can be derived from the electrical and thermal models. Particularly, coupling effects are incorporated into the thermal modeling of power devices. In this process, a Rainflow counting algorithm is employed to discern the number of temperature cycles with varying characteristics, such as the mean junction temperature T_{jm} and the junction temperature fluctuations ΔT_j . Subsequently, lifetime and damage accumulation

models for capacitors and power devices can be employed to estimate accumulated damage over a year. However, capacitance degradation affects capacitor reliability, and the feedback loop from the accumulated damage D_{mg} to the electrical model represents capacitance degradation. Due to the limited lifetime data, the lifetime in the block diagram is defined as B_{10} lifetime. This implies that only 10% of the samples will experience failure when the number of operating hours reaches this threshold. Once the accumulated damage surpasses 1, the component is regarded as having reached the end of its operational life. However, real-world situations introduce variations in the parameters of the component and lifetime models, affecting the probability distribution of failure. Consequently, Monte Carlo simulations based on a large number of samples need to be performed and the probability density functions (PDF) of each component can be derived by Weibull distribution fitting. Ultimately, the reliability block diagrams facilitate the determination of the PV converter's lifetime.

III. ELECTROTHERMAL MODELING OF CRITICAL COMPONENTS

A. Electrical Stress Analysis

Evaluating the influence of resonant currents on the lifetime of PV converters necessitates a preliminary analysis of the variation in electrical stresses within critical components, under different scenarios. Hence, assessments are conducted on the fundamental, resonant harmonic component (27th) and switching harmonic component (200th) of the current under rated power conditions. Fig. 5 shows the Fast Fourier Transform (FFT) simulation results of the inverter system in PLECS, comparing the filter capacitor currents and converter-side currents with and without resonant currents.

FFT analysis results of the capacitor current during converter stabilization and oscillation are shown in Fig. 5(a). When the damping method is employed, the capacitor current exhibits numerous switching harmonics in addition to the fundamental component, resulting in a THD of 172%. However, without resonant current, a substantial number of harmonic components, particularly around the 27th-order, appear in conjunction with switching harmonics, leading to a higher THD of 682%. Similarly, Fig. 5(b) presents FFT results for the converter-side current. Applying damping results in a THD of 14.5%, while the resonant component increases THD to 38.2% when the damping fails. This demonstrates that the resonant current amplifies electrical stress on the IGBT module and capacitors.

B. Experimental Validation

To expose the disparity in electrical stresses on the critical components with and without resonant currents, a single-phase *LCL* filter based power converter is constructed, as illustrated in Fig. 6, utilizing the parameters outlined in Table I. Measurements of grid voltages and currents are taken to observe the converter's operational conditions, while the currents passing through the IGBT modules and the capacitor are also measured to assess fluctuations in electrical stresses on the critical components.

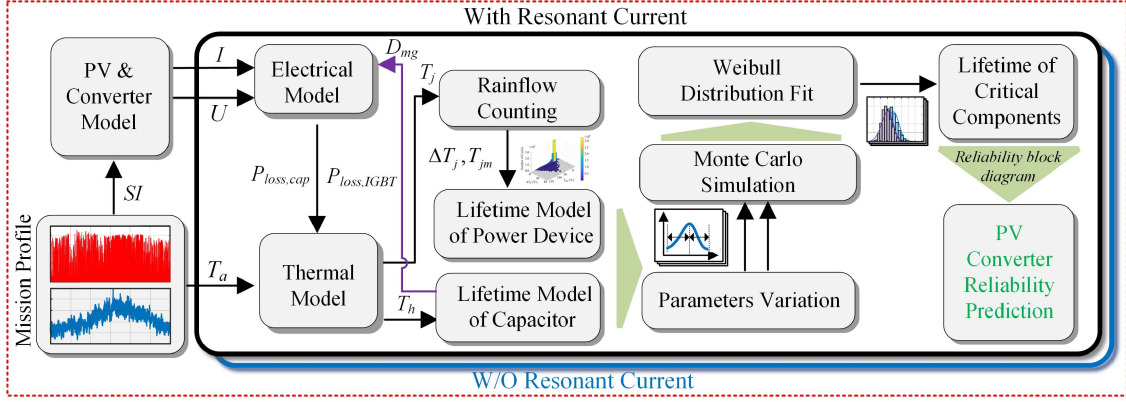


Fig. 4. Reliability assessment of PV converter based on mission profiles.

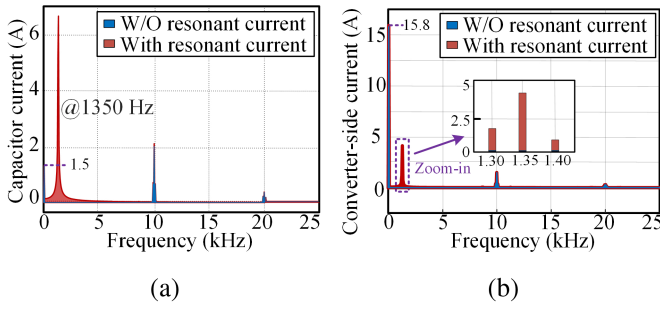
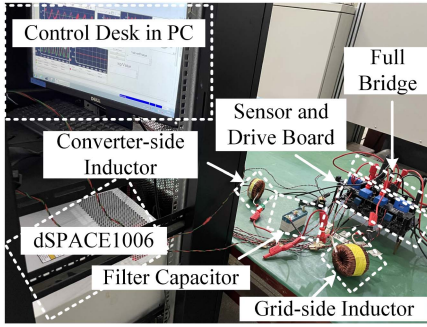


Fig. 5. FFT analysis of simulation results. (a) Capacitor current. (b) Converter-side current.


 Fig. 6. Experimental prototype of a single-phase *LCL*-filtered converter.

The converter's stability is evident in Fig. 7(a), where the application of the damping method effectively suppresses the resonance of the *LCL* filter. Notably, during converter stabilization, the switching harmonics of the capacitor current and converter-side current become prominent. These harmonics are concentrated around 10 kHz and 20 kHz, resulting in a THD of 175% and 16.1%, respectively, as illustrated in Fig. 8(a) and (c). Experimental waveforms for damping failure are depicted in Fig. 7(b), providing a clear illustration of converter oscillations. The FFT analysis of both capacitor currents and converter-side currents, as shown in Fig. 8(b) and (d), reveals the dominance of the 27th current harmonic along with 10 kHz and 20 kHz switching harmonics. The corresponding THD values are 686% and 40.2%, respectively. These findings are consistent with the simulated FFT analysis presented in Fig. 5. Hence, in the

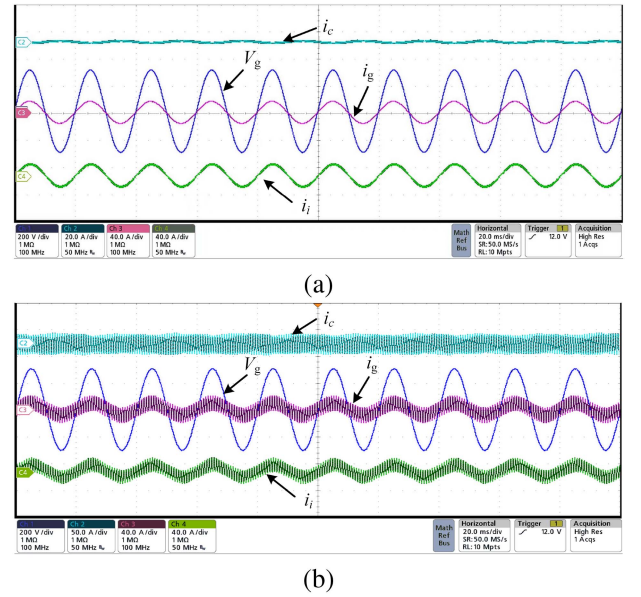


Fig. 7. Experimental results. (a) During converter stabilization. (Grid current i_g : 40 A/div, grid voltage V_g : 200 V/div, capacitor current i_c : 20 A/div, converter-side current i_L : 40 A/div, and time: 20 ms/div.) (b) During converter oscillations. (Grid current i_g : 40 A/div, grid voltage V_g : 200 V/div, capacitor current i_c : 50 A/div, converter-side current i_L : 40 A/div, and time: 20 ms/div.)

presence of resonance generated by the *LCL* filter, the electrical stresses on critical components are elevated, subsequently affecting their reliability. This will be quantitatively analyzed and depicted in the following sections.

C. Power Loss Modeling

1) *Capacitors*: To evaluate thermal stress for the selected inverters, it is crucial to execute a comprehensive analysis of power losses. The power loss of the MPF-CAP is shown in (5).

$$P_{loss,C} = \sum_n \left[ESR \times I_{rms}^2(f_n) + \frac{\tan \delta}{\omega_n C} \times I_{rms}^2(f_n) \right] \quad (5)$$

where ESR and $I_{rms}(f_n)$ represent the equivalent series resistance of the capacitor and the root-mean-square (rms) value of the ripple current at frequency f_n , respectively. n and δ denote

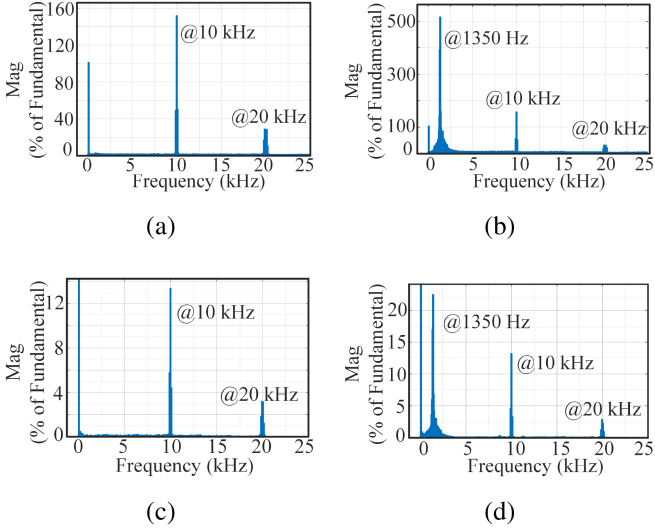


Fig. 8. FFT analysis results. (a) Capacitor current without resonant current. (b) Capacitor current with resonant current. (c) Converter-side current without resonant current. (d) Converter-side current with resonant current.

the harmonic order and dissipation factor, respectively. It is important to emphasize that the ESR curve of a capacitor is variable and dependent on the frequency. Thus, before analyzing the effect of harmonics on the thermal stress of the capacitor, it is necessary to convert the dominant harmonic frequencies to the corresponding values specified in the capacitor manufacturer's datasheet [11].

2) *Power Semiconductor Devices*: The loss dissipation of the power device consists mainly of the conduction loss and switching loss [12], which can be obtained using the following equations [41]:

$$P_{condT} = \frac{1}{T_0} \int_0^{T_0} (V_{ce0} i_T(t) + R_{ce} i_T^2(t)) \cdot D(t) dt \quad (6)$$

$$P_{condD} = \frac{1}{T_0} \int_0^{T_0} (V_{f0} i_D(t) + R_f i_D^2(t)) \cdot (1 - D(t)) dt \quad (7)$$

$$P_{swT} = \frac{1}{T_0} \int_0^{T_0} f_{sw} \cdot (a_T i_T^2(t) + b_T i_T(t) + c_T) \cdot \frac{U_{dc}}{U_{nom}} dt \quad (8)$$

$$P_{swD} = \frac{1}{T_0} \int_0^{T_0} f_{sw} \cdot (a_D i_D^2(t) + b_D i_D(t) + c_D) \cdot \frac{U_{dc}}{U_{nom}} dt \quad (9)$$

where $P_{condT/D}$ represents the conduction loss, $P_{swT/D}$ represents the switching loss, and T_0 is the fundamental period. V_{ce0} and V_{f0} are the initial voltage drops of the IGBT and the diode, while R_{ce} and R_f are the conduction resistances. $i_{T/D}$ is the current through each power component, f_{sw} is the switching frequency, U_{dc} is the dc-link voltage, and $D(t)$ denotes the duty cycle. The parameters $a_{T/D}$, $b_{T/D}$, and $c_{T/D}$ are determined based on the turn-on and turn-off energy dissipation of the IGBT, as well as the reverse recovery energy dissipation of the diode.

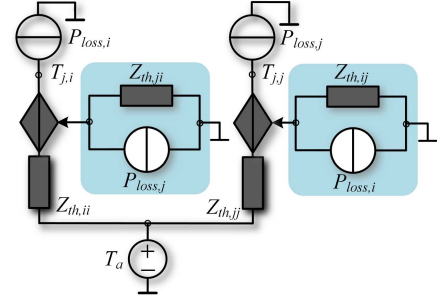


Fig. 9. Thermal model for the two components considering thermal coupling.

These values are typically provided by the manufacturer at a certain DC-link voltage U_{nom} .

D. Thermal Modeling

1) *Hotspot Temperature of Capacitor*: Thermal stress is a critical factor for capacitor failure, which can be quantified by the hotspot temperature T_h of the capacitor. Based on the mathematical model, taking the loading transient as an example, the hotspot temperature T_h of MPF-CAP is affected by the power loss $P_{loss,C}$ and ambient temperature T_a [29], as shown in (10).

$$T_h = \Delta T + T_a = P_{loss,C} R_{th} + T_a \quad (10)$$

where ΔT and R_{th} denote the hotspot temperature fluctuation and the equivalent thermal resistance, respectively.

2) *Junction Temperature of Power Device*: Resonant currents result in self-heating within the power module, intensifying thermal coupling effects in the selected chip. This increase in junction temperature accelerates the aging of the power devices, creating a significant risk to the overall reliability of the PV inverter system. In the thermal modeling of power devices, two commonly used types of thermal networks represent thermal behavior: the Cauer structure based on physical significance and the Foster structure based on experimental results. However, compact power device packaging is increasingly popular, but it introduces a notable challenge: the thermal cross-coupling effect between different semiconductor components. Fig. 9 illustrates the thermal impedance network connecting components i and j , comprising the coupling thermal impedances denoted as $Z_{th,ij}$ and $Z_{th,ji}$, respectively. By considering both self-heating and thermal cross-coupling effects and applying the superposition theory, the thermal model can be expressed in matrix form.

$$\begin{bmatrix} T_{j1}(t) \\ T_{j2}(t) \\ \vdots \\ T_{jN}(t) \end{bmatrix} = \frac{d}{dt} \begin{bmatrix} Z_{th,11}(t) & Z_{th,12}(t) & \cdots & Z_{th,1N}(t) \\ Z_{th,21}(t) & Z_{th,22}(t) & \cdots & Z_{th,2N}(t) \\ \vdots & \vdots & \ddots & \vdots \\ Z_{th,N1}(t) & Z_{th,N2}(t) & \cdots & Z_{th,NN}(t) \end{bmatrix} * \begin{bmatrix} P_{loss,1}(t) \\ P_{loss,2}(t) \\ \vdots \\ P_{loss,N}(t) \end{bmatrix} + T_a(t) \quad (11)$$

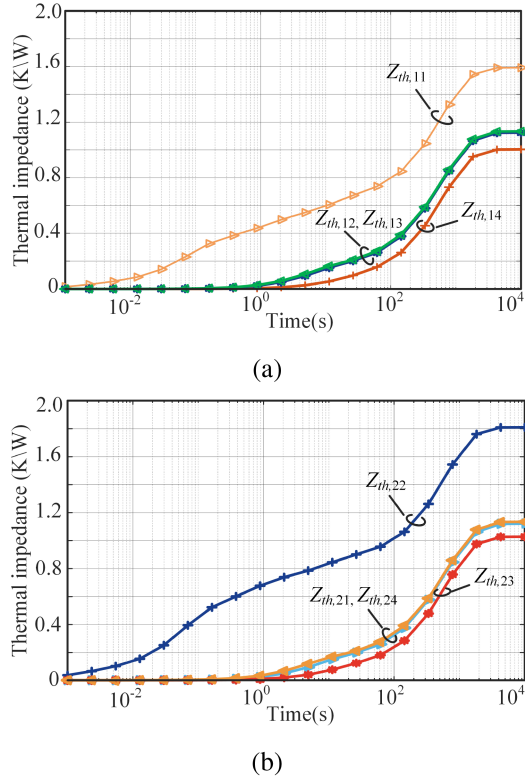


Fig. 10. Thermal impedance between components. (a) IGBT self-thermal impedance and coupling thermal impedance from junction to ambient. (b) Diode self-thermal impedance and coupling thermal impedance from junction to ambient.

where T_{jN} denotes the junction temperature of the N^{th} component, and the symbol “*” signifies convolution. Meanwhile, $Z_{th,NN}$ represents the thermal impedance from its self-junction to the ambient. Similarly, $Z_{th,Ni}$ denotes the thermal impedance from the mutual junction to the ambient between the i^{th} and N^{th} components. T_a corresponds to the ambient temperature, and $P_{loss,N}$ is the power loss of the N^{th} component. According to Infineon’s chip layout, the thermal coupling effect among more distant components is neglected, with the focus solely on extracting thermal coupling within the half-bridge. The self-thermal impedance and coupling impedance of the IGBT and diode are extracted and fitted by finite element method (FEM) simulation. To determine the coupled thermal impedance, the geometric configuration and material properties of the IGBT modules must first be entered into finite element simulation software. Then, a step power loss is applied to analyze the response curve of each chip. Simultaneously, junction temperatures are measured to assess the self-heating of individual chips and their thermal interaction with neighboring chips. Lastly, the thermal resistance between two nodes is calculated by dividing the measured temperature response by the applied power loss, as shown in Fig. 10.

Fig. 11 compares the effect of harmonic current on the capacitor hotspot temperature T_h and the power device junction temperature T_j with and without resonant currents. This temperature disparity can be primarily attributed to the substantial current harmonics generated by converter resonance, which flow through the critical components when the damping method fails.

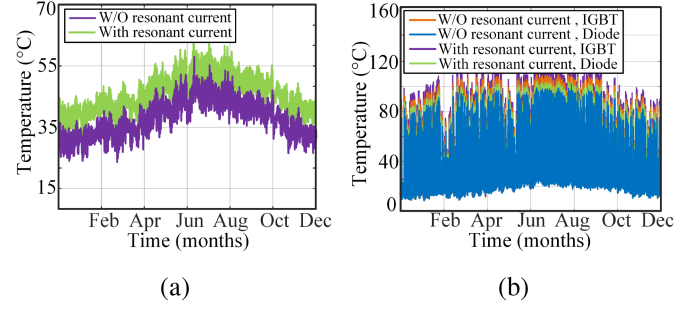


Fig. 11. Temperature profiles. (a) Capacitor hotspot temperature T_h . (b) Power device junction temperature T_j .

TABLE II
KEY PARAMETERS RELATED TO LIFETIME OF THE POWER MODULES

A	β_1	β_2	β_3
$1.42 \cdot 10^{12}$	-7.14	5154	-0.3

IV. LIFETIME ASSESSMENT OF PV CONVERTER

Power electronics reliability raises several open questions, spanning from the analysis of physical failure mechanisms to the validation of actual lifetimes. Consequently, it is crucial to note that the following assumptions are generally applicable to the different evaluated lifetime models.

- Damage accumulation follows Miner’s rule and accumulates in a linear manner [34], [37].
- The confidence level due to parameter variations is not considered in this paper [12].

A. Lifetime Modeling of Critical Components

The failure mechanism of power devices is frequently associated with bond wire lift-off, as well described in the literature [15]. In such cases, the evaluation of the lifetime is impacted by thermal cycling factors, such as the mean junction temperature T_{jm} and junction temperature fluctuations ΔT_j . However, the data mentioned above cannot be directly derived from the junction temperature profile. To obtain this information, Rainflow counting is employed to extract thermal cycles [20], enabling the mapping of these irregularities to the lifetime model based on empirical data. Regarding the lifetime modeling of power devices, the Bayerer’s model as shown in (12) is commonly employed [20].

$$N_f = A \cdot \Delta T_j^{\beta_1} \cdot e^{\left(\frac{\beta_2}{T_{jm} + 273}\right)} \cdot t_{on}^{\beta_3} \quad (12)$$

where the number of cycles to failure N_f is closely related to the rainflow results T_{jm} , ΔT_j , and t_{on} . Besides, A , β_1 , β_2 , and β_3 are from curve fitting of power cycle test data provided by the semiconductor manufacturer [20], [22], [42]. Table II provides key parameters of the capacitors used in the lifetime evaluation. It is important to note that these coefficients are typically derived by manufacturers through accelerated testing. However, the specific conditions of these tests and the correlation between the resulting coefficients can limit the model’s accuracy and applicability. In this paper, to further investigate the unclear

TABLE III
KEY PARAMETERS RELATED TO LIFETIME OF MPF-CAPS

Parameters	Values
Rated voltage V_0 (V)	330
Rated temperature T_0 ($^{\circ}\text{C}$)	70
Voltage stress exponent n_2	0.012
Dissipation factor $\tan \delta$	6e-4
Rated lifetime L_0 (hour)	30000

thermal behavior of LCL -type inverter systems influenced by resonant currents, the common interpolation method is applied to some of the data [43]. Consequently, more advanced lifetime models are required to account for a broader range of loading conditions in power devices, aiming to enhance the precision of lifetime estimation under complex operating conditions.

A widely used capacitor lifetime model [18] with respect to hotspot temperature and voltage can be expressed as

$$L_x = L_0 \times 2^{\frac{T_0 - T_h}{10}} \times (V_x/V_0)^{-n_2} \quad (13)$$

where L_0 , V_0 , and T_0 denote the rated lifetime, rated voltage, and rated temperature, respectively. L_x , V_x , T_h , and n_2 represent the calculated lifetime, capacitor operating voltage, hotspot temperature, and voltage stress exponent, respectively. These parameters used for capacitor lifetime prediction are listed in Table III.

When assessing the long-term reliability of a PV inverter based on its mission profile, it is essential to consider the impact of capacitance degradation on the converter's lifetime. Typically, accumulated damage can be converted to capacitance loss to characterize the impact of capacitance degradation during the assessment process [44].

$$\Delta C(t) = \sum_{i=1} \frac{t_i}{L(i)} \cdot 5\%C \quad (14)$$

where $\Delta C(t)$ represents the capacitance loss of a film capacitor at the operating time t_i , and $L(i)$ denotes the expected lifetime under specific operating condition i , existing research has established that the end of a film capacitor's lifetime is defined by a 5% loss in its initial capacitance [45]. However, due to insufficient test data, an accurate exponential model for predicting capacitance loss is not available. For capacitance losses below 5%, the degradation can be approximated as linear.

Considering the Miner's rule, the accumulated damage of power devices is calculated by

$$D_{sw} = \sum_{i=1} \frac{n_i}{(N_f)_i} \quad (15)$$

where n_i is the number of cycles under a specific thermal stress and $(N_f)_i$ is the number of cycles until failure under the same stress. When D_{sw} exceeds 1, the component is considered to have reached its end of life. Observations gleaned from Fig. 12(a) and (b) reveal a persistent trend: in the case where the damping method is ineffective in suppressing the resonance, resonant current results in elevated accumulated damage, contrasting with the mitigating effect of the damping method. Furthermore, IGBTs lead to substantial damage of up to 3.02% under resonance,

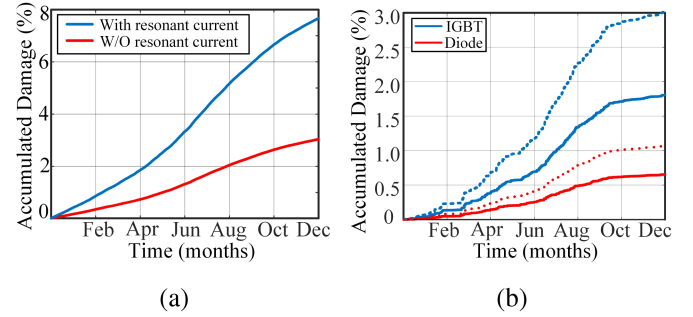


Fig. 12. Annual damage of critical components with and without resonant currents. (a) Capacitor. (b) Power devices (solid line: without resonant current; dotted line: with resonant current).

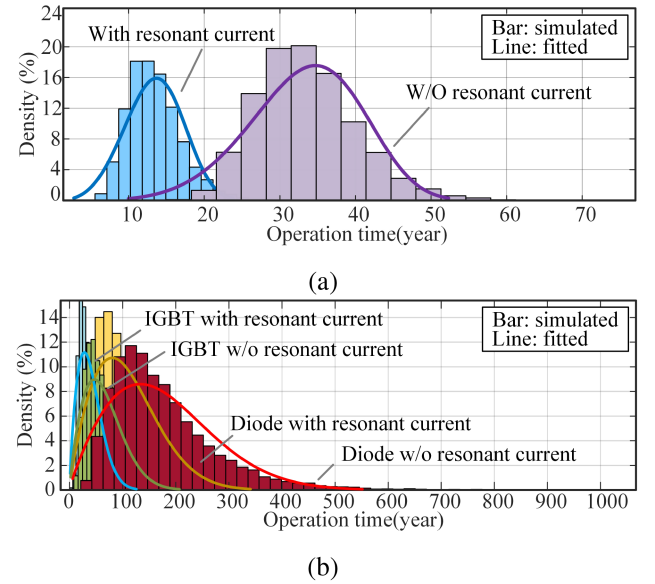


Fig. 13. The Weibull distribution. (a) Histogram of years to failure of capacitors with and without resonant currents. (b) Histogram of years to failure of power devices with and without resonant currents.

which is nearly three times higher than the 1.07% damage observed in diodes. The strong electrical stress from resonant currents results in capacitor damage of up to 7.6%.

B. Monte Carlo Simulation and Lifetime Prediction

In real-world applications, the lifetime of critical components is influenced by multiple uncertainties, particularly those associated with parameters within the lifetime model. To enhance the precision of the converter reliability assessment, the lifetime assessment process integrates Monte Carlo simulation to address these uncertainties. Assuming that all parameters in the lifetime model conform to a standard normal distribution, a fixed value of standard deviation is applied, e.g., 5% in [22], [42]. Fig. 13 shows the histograms obtained from Monte Carlo simulations performed with a total of 10,000 samples. Typically, the histograms of years to failure over time follow the Weibull distribution, i.e.,

$$f(t) = (\beta/\eta) \cdot (t/\eta)^{\beta-1} \cdot e^{-(t/\eta)^\beta}, F(t) = 1 - e^{-(t/\eta)^\beta} \quad (16)$$

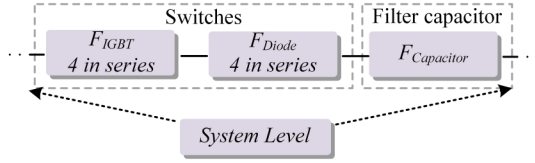


Fig. 14. Converter-level reliability block diagram consisting of the power devices and capacitor.

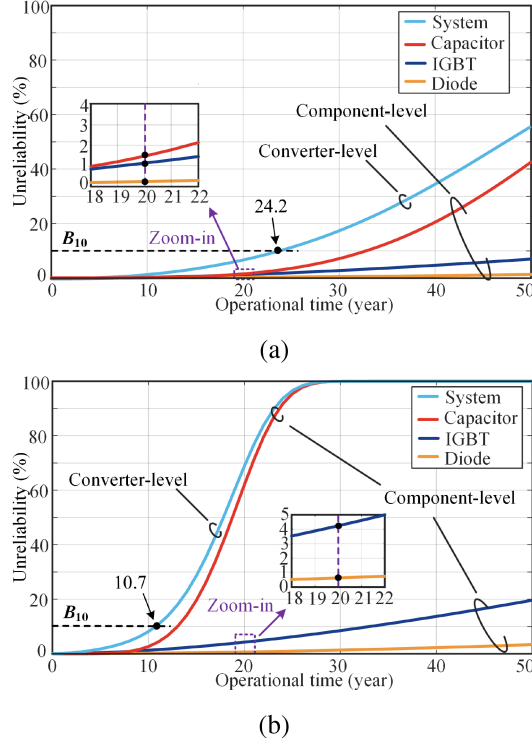


Fig. 15. Unreliability curve from a single component to the PV converter. (a) Without resonant current. (b) With resonant current.

where $f(t)$ and $F(t)$, respectively, represent the PDF of the Weibull distribution and the unreliability function. β denotes the shape parameter, and η denotes the scale parameter.

Regarding the evaluation of converter-level reliability, as explained in the preceding flowchart, the reliability block diagram can be utilized to depict the reliability interactions among individual components within the converter. If there is no parallel path in the reliability block diagram, the reliability of the converter is the multiplication of all components inside, as illustrated in Fig. 14. The unreliability of the whole converter $F_{sys}(t)$ can be expressed as

$$F_{sys}(t) = 1 - \prod_j (1 - F_{components,j}(t)) \quad (17)$$

where $F_{components,j}$ denotes the unreliability of the j -th component in the converter. It is evident that the converter-level reliability may be weakened with an increasing number of components.

Fig. 15 illustrates the probability of failure curves for critical components and the converter, both with and without resonant currents. Notably, the resonant current significantly influences converter failure. With the application of damping methods, the

TABLE IV
PARAMETERS AND RESONANT FREQUENCY SETTINGS IN THE CASE SETS

Settings	Case 1	Case 2	Case 3
C_f	17 μ F	16 μ F	15 μ F
f_{res}	1254 Hz	1293 Hz	1335 Hz

probability of failure for critical components before 20-years of operation is considerably lower, especially for diodes with the lowest failure probability. If the damping method operates ineffectively, the failure probabilities of critical components before 20 years of operation are 62.08% (capacitor), 4.24% (IGBT), and 0.65% (diode), indicating that the resonant currents result in a substantial exponential underestimation of the failure probability. However, the B_{10} lifetime of the PV converter, with and without considering resonant currents, is 10.7 and 24.2 years, respectively. This implies that the presence of resonant currents reduces the lifetime by about 55.5%. Additionally, the effective application of the damping method guarantees that the converter meets its designed 20 years lifetime expectation [42]. Similarly, as depicted in Fig. 15, it is evident that the probability of failure for the filter capacitor dominates the converter's failure. To enhance the reliability of the PV inverter converter, the primary focus should be on reducing the failure probability of the capacitor.

V. DISCUSSIONS

To enhance the practical relevance of the preceding reliability analysis, this section focuses on two critical dynamic factors influencing the long-term performance of *LCL*-type PV converters: resonant frequency and harmonic content level. These factors are both prevalent and variable in real-world operating conditions, exerting a significant influence on component aging and system degradation. The objective is to bridge the gap between theoretical modeling and field reliability, offering a reliability-oriented perspective for *LCL*-type converter design.

A. Effect of Resonant Frequency on Reliability

Without loss of generality, this paper assumes that inductance values remain relatively stable during operation, since magnetic components typically exhibit low parameter drift. In contrast, filter capacitors are more susceptible to degradation due to aging, resulting in a decrease in capacitance over time. Therefore, this study considers only the effects of capacitor degradation on the system's resonant characteristics. To reflect practical degradation scenarios in *LCL* filters, it is assumed a 20% reduction in the filter capacitor value, which is within the commonly observed aging range. Based on this variation, three representative resonant frequency cases are selected for comparative analysis. The specific parameters and resonant frequencies are summarized in Table IV.

Resonant frequency is a key design parameter that influences both control stability and the distribution of current stress among critical components in the converter. In this study, the impact of

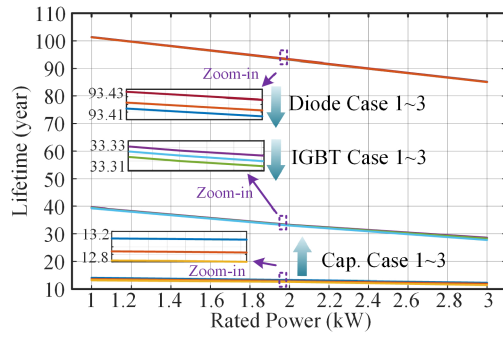


Fig. 16. Lifetime evaluation results with different power ratings in case 1 to 3.

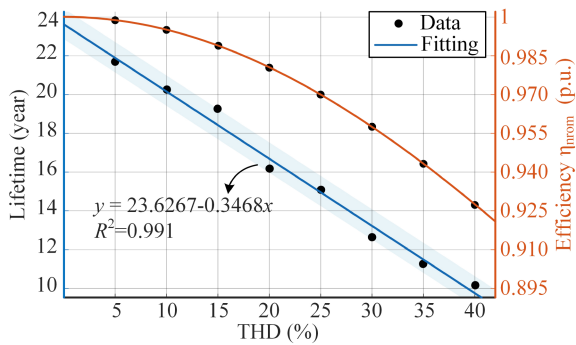


Fig. 17. Impact of total harmonic distortion on system output efficiency and operational lifetime. ($\eta_{\text{norm}} \approx 1/\sqrt{1 + \text{THD}^2}$).

different resonant frequencies on system reliability is quantitatively evaluated using the previously developed electro-thermal model. As shown in Fig. 16, the results reveal a clear trend: as the resonant frequency increases, the reliability of power devices decreases slightly, while the reliability of filter capacitors improves significantly. This is because with higher resonant frequency, the negligible increase in power losses within the power device, whereas the losses in the filter capacitors are substantially reduced. Moreover, according to Fig. 16, the greater the rated power, the greater the impact of the resonant current on the lifetime. These findings suggest that the selection of filter parameters should consider the potential to enhance converter lifetime, especially under conditions where parameter drift is expected.

B. Impact of Different Resonant Contents on Reliability

Under non-ideal grid conditions or system disturbances, the output current of *LCL*-type converters often contains significant harmonic content. The varying resonant current content results in differences in power loss, junction temperature, and hotspot temperature. In this subsection, the effect of THD on the inverter system's lifetime is analyzed, with an upper limit of 40% THD. Fig. 17 illustrates the relationship between THD levels, converter lifetime, and inverter output efficiency. The results indicate that the converter's lifetime decreases approximately linearly with increasing THD. This trend aligns with prior theoretical analyses, which suggest that resonant currents impose elevated

electrical and thermal stresses on critical components, thereby accelerating degradation and failure. Additionally, resonance-induced high-frequency components contribute to increased power losses, leading to a measurable decline in efficiency. Using the evaluation method proposed in this paper, an approximately linear relationship between THD and converter lifetime is observed. According to the fitted curves, under these specific conditions, the converter's reliability will meet its expected design lifetime as long as the THD remains below 10% when resonant currents are present. The establishment of quantitative relationships provides a framework for visualizing the effects of varying resonant current levels on converter reliability and universalizes the evaluation process.

Based on the above discussions, a reliability-oriented design framework is proposed for *LCL*-type PV inverters. This framework systematically integrates considerations of resonant frequency and harmonic robustness into the design process, enabling quantitative evaluation of the relationship between converter lifetime and varying operational factors. The findings highlight the importance of not only satisfying harmonic performance criteria but also adapting filter design to sustain long-term system reliability under resonant current conditions, particularly in the case of capacitance parameter drift.

VI. CONCLUSION

This paper presents a comprehensive reliability evaluation framework for *LCL*-type PV inverters. Initially, a quantitative assessment is conducted to evaluate the impact of resonant currents on the reliability of *LCL*-type PV inverters. This is achieved through detailed modeling of the electrical, thermal, and physical characteristics of key components, combined with lifetime prediction techniques. In addition, a comparative analysis of component lifetimes is performed with respect to resonant frequency and harmonic content, aiming to optimize *LCL*-type PV converter design from a reliability perspective. Based on these case studies, the following conclusions can be drawn:

- 1) A detailed quantitative assessment of the impact of resonant current on component degradation is performed, revealing its dominant role in accelerating the failure of capacitors and power devices.
- 2) The electro-thermal model explicitly incorporates IGBT thermal coupling and capacitor aging due to capacitance derating, allowing a realistic prediction of lifetime under long-term operational conditions.
- 3) Power loss and thermal profiles obtained under both resonant and non-resonant conditions are combined with mission profiles and Monte Carlo-based lifetime estimations using Weibull statistics. The results indicate that improper damping control or parameter drift can induce significant resonant stress, reducing the system lifetime by more than 55% (from 24.2 years to 10.7 years). Moreover, the reliability of the filter capacitor is identified as the dominant factor influencing overall system longevity.
- 4) A quantitative relationship between resonant frequency, harmonic content and system reliability is established. The results show that increasing the resonant frequency will

improve the lifetime of the capacitor, while reducing the THD significantly improves the device's stress margin.

In practice, a reliability-oriented design framework is proposed in the presence of resonant currents to guide the design of *LCL*-type PV converters. The framework combines reliability analysis with filter characterization to achieve long-term reliable operation of *LCL*-type PV converters under realistic grid conditions.

REFERENCES

- [1] J. Sun, X. Zhang, T. Kerekes, W. Yao, X. Zhang, and D. Zhou, "Resonant current impacts on the lifetime of LCL filter in PV systems," in *Proc. IEEE Energy Convers. Congr. Expo.*, 2023, pp. 1360–1365.
- [2] F. Blaabjerg, Y. Yang, D. Yang, and X. Wang, "Distributed power-generation systems and protection," *Proc. IEEE*, vol. 105, no. 7, pp. 1311–1331, Jul. 2017.
- [3] H. Wang and F. Blaabjerg, "Power electronics reliability: State of the art and outlook," *IEEE J. Emerg. Sel. Topics Power Electron.*, vol. 9, no. 6, pp. 6476–6493, Dec. 2021.
- [4] X. Wang, F. Blaabjerg, and P. C. Loh, "Passivity-based stability analysis and damping injection for multiparalleled VSCs with *LCL* filters," *IEEE Trans. Power Electron.*, vol. 32, no. 11, pp. 8922–8935, Nov. 2017.
- [5] M. Liserre, F. Blaabjerg, and S. Hansen, "Design and control of an *LCL*-filter-based three-phase active rectifier," *IEEE Trans. Ind. Appl.*, vol. 41, no. 5, pp. 1281–1291, Sep./Oct. 2005.
- [6] A. Sangwongwanich, Y. Yang, D. Sera, and F. Blaabjerg, "Lifetime evaluation of grid-connected PV inverters considering panel degradation rates and installation sites," *IEEE Trans. Power Electron.*, vol. 33, no. 2, pp. 1225–1236, Feb. 2018.
- [7] J. He, A. Sangwongwanich, Y. Yang, and F. Iannuzzo, "Lifetime evaluation of three-level inverters for 1500-V photovoltaic systems," *IEEE J. Emerg. Sel. Topics Power Electron.*, vol. 9, no. 4, pp. 4285–4298, Aug. 2021.
- [8] L. M. Moore and H. N. Post, "Five years of operating experience at a large utility-scale photovoltaic generating plant," *Prog. Photovolt. Res. Appl.*, vol. 16, no. 3, pp. 249–259, 2008.
- [9] H. Wang et al., "Transitioning to physics-of-failure as a reliability driver in power electronics," *IEEE J. Emerg. Sel. Topics Power Electron.*, vol. 2, no. 1, pp. 97–114, Mar. 2014.
- [10] P. Hacke et al., "A status review of photovoltaic power conversion equipment reliability, safety, and quality assurance protocols," *Renewable Sustain. Energy Rev.*, vol. 82, pp. 1097–1112, 2018.
- [11] D. Zhou, H. Wang, and F. Blaabjerg, "Mission profile based system-level reliability analysis of DC/DC converters for a backup power application," *IEEE Trans. Power Electron.*, vol. 33, no. 9, pp. 8030–8039, Sep. 2018.
- [12] D. Zhou, F. Blaabjerg, T. Franke, M. Tønnes, and M. Lau, "Comparison of wind power converter reliability with low-speed and medium-speed permanent-magnet synchronous generators," *IEEE Trans. Ind. Electron.*, vol. 62, no. 10, pp. 6575–6584, Oct. 2015.
- [13] D. Zhou, G. Zhang, and F. Blaabjerg, "Optimal selection of power converter in DFIG wind turbine with enhanced system-level reliability," *IEEE Trans. Ind. Appl.*, vol. 54, no. 4, pp. 3637–3644, Jul./Aug. 2018.
- [14] H. Wang and F. Blaabjerg, "Reliability of capacitors for DC-link applications in power electronic converters—an overview," *IEEE Trans. Ind. Appl.*, vol. 50, no. 5, pp. 3569–3578, Sep./Oct. 2014.
- [15] C. Zhan, L. Zhu, W. Wang, Y. Zhang, S. Ji, and F. Iannuzzo, "Multidimensional mission-profile-based lifetime estimation approach for IGBT modules in MMC–HVDC application considering bidirectional power transfer," *IEEE Trans. Ind. Electron.*, vol. 70, no. 7, pp. 7290–7300, Jul. 2023.
- [16] B. Yao, X. Ge, H. Wang, H. Wang, D. Zhou, and B. Gou, "Multiscale reliability evaluation of DC-link capacitor banks in metro traction drive system," *IEEE Trans. Transp. Electrification*, vol. 6, no. 1, pp. 213–227, Mar. 2020.
- [17] X. Pei, W. Zhou, and Y. Kang, "Analysis and calculation of DC-link current and voltage ripples for three-phase inverter with unbalanced load," *IEEE Trans. Power Electron.*, vol. 30, no. 10, pp. 5401–5412, Oct. 2015.
- [18] Z. Zhao, D. Zhou, P. Davari, J. Fang, and F. Blaabjerg, "Reliability analysis of capacitors in voltage regulator modules with consecutive load transients," *IEEE Trans. Power Electron.*, vol. 36, no. 3, pp. 2481–2487, Mar. 2021.
- [19] D. Zhou, H. Wang, and F. Blaabjerg, "Reactive power impacts on *LCL* filter capacitor lifetime in grid-connected inverter," *IEEE Open J. Power Electron.*, vol. 1, pp. 139–148, 2020.
- [20] S. Ye, D. Zhou, X. Yao, and F. Blaabjerg, "Component-level reliability assessment of a direct-drive PMSG wind power converter considering two terms of thermal cycles and the parameter sensitivity analysis," *IEEE Trans. Power Electron.*, vol. 36, no. 9, pp. 10037–10050, Sep. 2021.
- [21] A. Antonopoulos, S. D'Arco, M. Hernes, and D. Pefitsis, "Limitations and guidelines for damage estimation based on lifetime models for high-power IGBTs in realistic application conditions," *IEEE J. Emerg. Sel. Topics Power Electron.*, vol. 9, no. 3, pp. 3598–3609, Jun. 2021.
- [22] D. Zhou and F. Blaabjerg, "Converter-level reliability of wind turbine with low sample rate mission profile," *IEEE Trans. Ind. Appl.*, vol. 56, no. 3, pp. 2938–2944, May/Jun. 2020.
- [23] Y. Song and B. Wang, "Survey on reliability of power electronic systems," *IEEE Trans. Power Electron.*, vol. 28, no. 1, pp. 591–604, Jan. 2013.
- [24] T. Dragičević, P. Wheeler, and F. Blaabjerg, "Artificial intelligence aided automated design for reliability of power electronic systems," *IEEE Trans. Power Electron.*, vol. 34, no. 8, pp. 7161–7171, Aug. 2019.
- [25] Polypropylene-phb-rhb. Icel S.r.l. Corp., Castellanza, Italy, 2022. [Online]. Available: https://www.icel.it/docs/4022/polypropylene_phb_rhb_2ae5325f58.pdf
- [26] W. Yao, Y. Yang, Y. Xu, F. Blaabjerg, S. Liu, and G. Wilson, "Phase reshaping via all-pass filters for robust *LCL*-filter active damping," *IEEE Trans. Power Electron.*, vol. 35, no. 3, pp. 3114–3126, Mar. 2020.
- [27] X. Wang, F. Blaabjerg, and P. C. Loh, "Grid-current-feedback active damping for *LCL* resonance in grid-connected voltage-source converters," *IEEE Trans. Power Electron.*, vol. 31, no. 1, pp. 213–223, Jan. 2016.
- [28] X. Wang, Y. He, D. Pan, H. Zhang, Y. Ma, and X. Ruan, "Harmonic instability of *LCL*-type grid-connected inverter caused by the pole-zero cancellation: A case study," *IEEE Trans. Ind. Electron.*, vol. 69, no. 11, pp. 11580–11589, Nov. 2022.
- [29] J. Sun, X. Zhang, W. Yao, X. Zhang, and D. Zhou, "Analysis of resonant current influence on capacitor lifetime variation in *LCL*-type photovoltaic systems," *IEEE Trans. Power Electron.*, vol. 39, no. 12, pp. 15682–15692, Dec. 2024.
- [30] W. Yao, Y. Yang, X. Zhang, F. Blaabjerg, and P. C. Loh, "Design and analysis of robust active damping for *LCL* filters using digital notch filters," *IEEE Trans. Power Electron.*, vol. 32, no. 3, pp. 2360–2375, Mar. 2017.
- [31] B. Yao et al., "Electrothermal stress analysis and lifetime evaluation of DC-link capacitor banks in the railway traction drive system," *IEEE J. Emerg. Sel. Topics Power Electron.*, vol. 9, no. 4, pp. 4269–4284, Aug. 2021.
- [32] D. Zhou, Y. Song, Y. Liu, and F. Blaabjerg, "Mission profile based reliability evaluation of capacitor banks in wind power converters," *IEEE Trans. Power Electron.*, vol. 34, no. 5, pp. 4665–4677, May 2019.
- [33] N. Agarwal, A. Arya, M. W. Ahmad, and S. Anand, "Lifetime monitoring of electrolytic capacitor to maximize earnings from grid-feeding PV system," *IEEE Trans. Ind. Electron.*, vol. 63, no. 11, pp. 7049–7058, Nov. 2016.
- [34] K. Ma, M. Liserre, F. Blaabjerg, and T. Kerekes, "Thermal loading and lifetime estimation for power device considering mission profiles in wind power converter," *IEEE Trans. Power Electron.*, vol. 30, no. 2, pp. 590–602, Feb. 2015.
- [35] P. D. Reigosa, H. Wang, Y. Yang, and F. Blaabjerg, "Prediction of bond wire fatigue of IGBTs in a PV inverter under a long-term operation," *IEEE Trans. Power Electron.*, vol. 31, no. 10, pp. 7171–7182, Oct. 2016.
- [36] A. Sangwongwanich, Y. Yang, D. Sera, and F. Blaabjerg, "Mission profile-oriented control for reliability and lifetime of photovoltaic inverters," *IEEE Trans. Ind. Appl.*, vol. 56, no. 1, pp. 601–610, Jan./Feb. 2020.
- [37] A. Sangwongwanich, Y. Yang, D. Sera, F. Blaabjerg, and D. Zhou, "On the impacts of PV array sizing on the inverter reliability and lifetime," *IEEE Trans. Ind. Appl.*, vol. 54, no. 4, pp. 3656–3667, Jul./Aug. 2018.
- [38] X. Li, J. Fang, Y. Tang, X. Wu, and Y. Geng, "Capacitor-voltage feed-forward with full delay compensation to improve weak grids adaptability of *LCL*-filtered grid-connected converters for distributed generation systems," *IEEE Trans. Power Electron.*, vol. 33, no. 1, pp. 749–764, Jan. 2018.
- [39] X. Wang, X. Ruan, S. Liu, and C. K. Tse, "Full feedforward of grid voltage for grid-connected inverter with *LCL* filter to suppress current distortion due to grid voltage harmonics," *IEEE Trans. Power Electron.*, vol. 25, no. 12, pp. 3119–3127, Dec. 2010.
- [40] X. Li, X. Wu, Y. Geng, X. Yuan, C. Xia, and X. Zhang, "Wide damping region for *LCL*-type grid-connected inverter with an improved capacitor-current-feedback method," *IEEE Trans. Power Electron.*, vol. 30, no. 9, pp. 5247–5259, Sep. 2015.

- [41] ABB, *Applying IGBTs*. Application Notes, 2009.
- [42] Y. Shen, A. Chub, H. Wang, D. Vinnikov, E. Liivik, and F. Blaabjerg, "Wear-out failure analysis of an impedance-source PV microinverter based on system-level electrothermal modeling," *IEEE Trans. Ind. Electron.*, vol. 66, no. 5, pp. 3914–3927, May 2019.
- [43] J. Sun, W. Yao, X. Zhang, X. Zhang, and D. Zhou, "Reliability assessment of IGBT modules in *LC*L-type photovoltaic systems considering resonant currents," *IEEE J. Emerg. Sel. Topics Power Electron.*, vol. 12, no. 6, pp. 5794–5804, Dec. 2024.
- [44] J. Wei, H. Feng, and L. Ran, "Multidimensional design of DC-link for two-stage solid-state transformer cell considering second-order harmonic current distribution," *IEEE Trans. Power Electron.*, vol. 38, no. 8, pp. 10244–10255, Aug. 2023.
- [45] Z. Li et al., "Lifetime prediction of metallized film capacitors based on capacitance loss," *IEEE Trans. Plasma Sci.*, vol. 41, no. 5, pp. 1313–1318, May 2013.



Xinyue Zhang (Member, IEEE) received the B.S., M.S., and Ph.D. degrees in electrical engineering from Northwestern Polytechnical University, Xi'an, China, in 2016, 2019, and 2025, respectively. She was a Visiting Student with Aalborg University, Aalborg, Denmark, from 2022 to 2023. Her current research interests include the thermal analysis and reliability of power electronic devices.



Jiacheng Sun (Member, IEEE) received the B.S., M.S., and Ph.D. degrees in electrical engineering from Northwestern Polytechnical University, Xi'an, China, in 2016, 2019, and 2025, respectively. From 2021 to 2023, he was a Visiting Ph.D. with Aalborg University, Aalborg, Denmark. He is currently working with Xi'an Thermal Power Research Institute Company, LTD, China. His research interests include current control, grid-connected inverters, reliability of power electronics in renewable energy application.



Zhongzheng Zhou (Member, IEEE) was born in Shandong, China, in 1989. He received the B.S. degree in electrical engineering and automation from the China University of Mining and Technology, Xuzhou, China, in 2014, and the Ph.D. degree in electrical engineering from Shanghai Jiao Tong University, Shanghai, China, in 2020. In 2018, from Feb. to Mar. he was a Guest Researcher with the University of Kaiserslautern, Kaiserslautern, Germany. He is currently an Associate Professor with the Department of Electrical Engineering, Northwestern Polytechnical

University. His research interests include dc circuit breakers, multilevel converters, and high-power motor drives.



Zhen Kang (Member, IEEE) was born in Shanxi, China, in 1988. He received the B.Sc. and Ph.D degrees of electrical engineering from Xi'an Jiaotong University, Xi'an, China, in 2010 and 2016, respectively. He is currently working with Xi'an Thermal Power Research Institute Company, LTD, China. His research interests are in the areas of condition monitoring and fault diagnosis of large generators.



Wenli Yao (Member, IEEE) received the B.S., M.S., and Ph.D. degrees in electrical engineering from Northwestern Polytechnical University, Xi'an, China, in 2009, 2012, and 2017, respectively. Since 2019, he has been with Northwestern Polytechnical University, Xi'an, China, as an Associate Professor. His research interests include current control, grid-connected inverter, multi-pulse converter, and power decoupling.



on topologies and control of power electronic converters.

Tamas Kerekes (Senior Member, IEEE) received the Diploma degree in electrical engineer with specialization in electric drives and robots from the Technical University of Cluj, Romania, in 2002, the Master of Science degree in power electronics and drives and the Ph.D. degree from the Institute of Energy Technology, Aalborg University. His Ph.D. program was titled: "Analysis and modeling of transformerless PV inverter systems". He is currently an Associate Professor with Aalborg University. His research interests include renewable energy systems with focus



Xiaobin Zhang received the B.S. and M.S degrees in electrical engineering from the School of Automation, Northwestern Polytechnical University, Xi'an, China, in 1983 and 1986, respectively. Since 1986, he has been with the School of Automation, Northwestern Polytechnical University, where he is currently a Professor. His research interests include power electronic control, aircraft power system, multi-pulse converter.



He was as an Associate Editor for IEEE TRANSACTIONS ON INDUSTRY APPLICATIONS. He was the recipient of the few IEEE prized paper awards.

Dao Zhou (Senior Member, IEEE) received the B.S. degree in electrical engineering from Beijing Jiaotong University, Beijing, China, in 2007, the M.S. degree in electrical engineering from Zhejiang University, Hangzhou, China, in 2010, and the Ph.D. degree in electrical engineering from Aalborg University, Aalborg, Denmark, in 2014. Since 2014, he has been with the Department of Energy, Aalborg University, where he is currently an Associate Professor. His research interests include modeling, control, and reliability of power electronics in renewable energy applications.